

6(a)(i)	1 <u>blood glucose concentration</u> increases until 30 minutes ; 2 decreases to original value / returns to 5 mmol dm ⁻³ , by 90 minutes ;	2
6(a)(ii)	83.3 ; ;	2
6(a)(iii)	any two from: 1 reduce intake of (named) carbohydrate ; 2 physical exercise ; 3 inject insulin ; 4 AVP ;	2

Question	Answer	Marks
6(b)	any seven from : 1 glucagon is, cell-signalling molecule / ligand / first messenger ; 2 binds to receptor on liver cell (surface) membrane ; 3 G-protein activated ; 4 aden(yl)yl cyclase activated / formation of cAMP ; 5 <i>ref. to</i> second messenger ; 6 enzyme cascade / signal amplified ; 7 glycogenolysis / glycogen hydrolysed to glucose ; 8 gluconeogenesis / amino acids → glucose / triglycerides → glucose ; 9 glucose released into blood ; 10 AVP ;	7

Question	Answer	Marks
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